



Data Analytics

Project 1

Industrial Training Kit

Batch: 2026 | Powered by DecodeLabs

WELCOME TO THE TEAM! 🚀

Step into the role of a Data Analyst at DecodeLabs. Project 1 is your foundation phase: Data Cleaning & Preparation. This track isn't about "fancy charts"—it's about Data Integrity. Before you build a single predictive model or dashboard, you must master the art of scrubbing raw, messy datasets into reliable sources of truth. By completing this milestone, you are proving you can identify missing values, eliminate duplicates, and correct formatting through pure analytical logic. Let's prepare a digital foundation that ensures our insights are built on absolute accuracy.



GUIDELINES:

Qualification Criteria

EARN YOUR BADGE 

- Requirement: Complete project 1 to unlock other projects for next week.
- Standard: All projects must be verified for quality.
- Mandatory: Project 1 is the essential start for every intern.



Project: 1

Data Cleaning & Preparation 🖌️



Goal:

Clean a raw dataset by handling missing values, duplicates, and incorrect data.

Key Requirements:

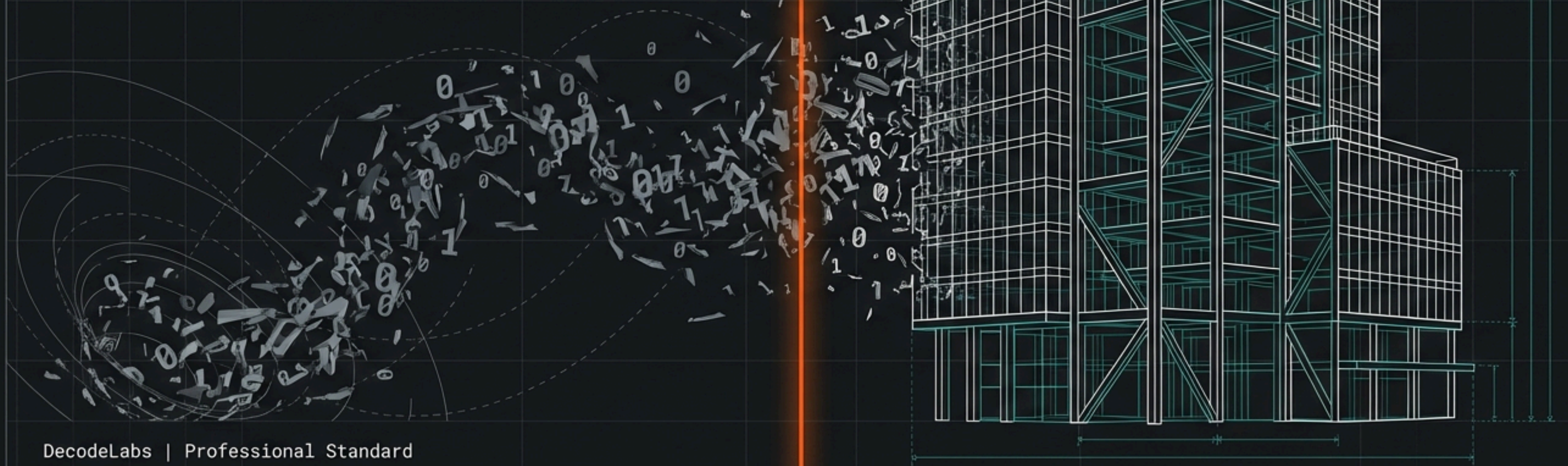
- Identify missing or null values
 - Remove duplicates
- Correct data formats (dates, numbers, text)

Key Skills:

Data cleaning, Excel or Python basics, data preparation

The Architecture of Data Integrity

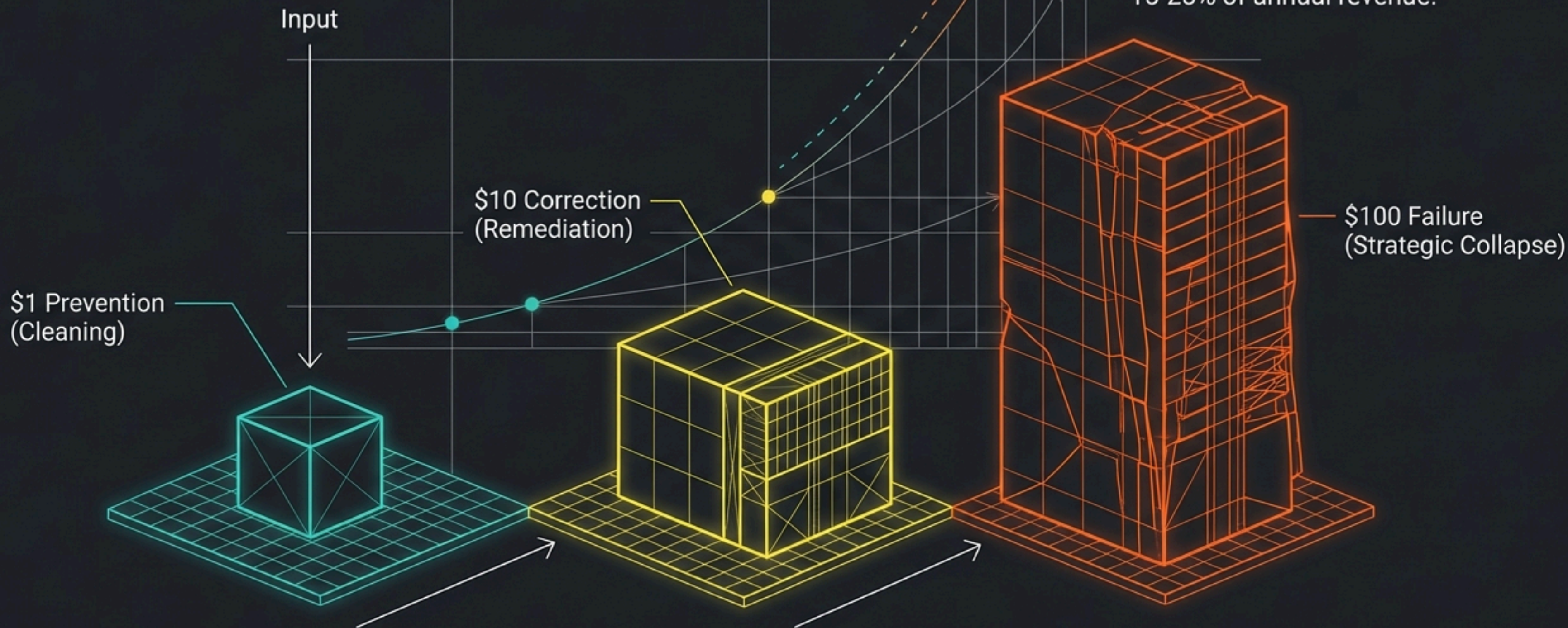
Intern Project 1: Transforming Raw Noise into Business Intelligence.



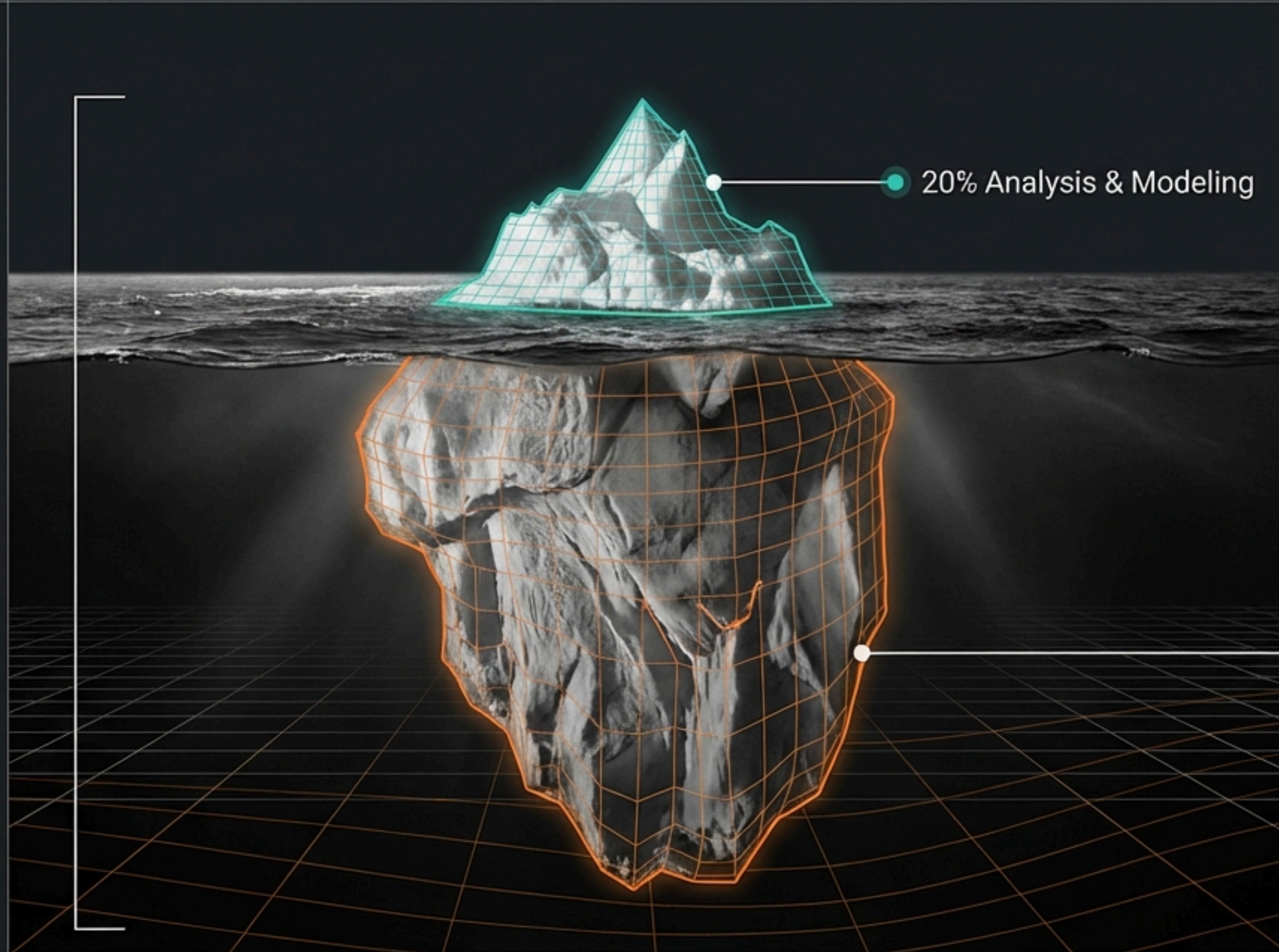
THE EXPONENTIAL COST OF ERROR

AVG. ANNUAL LOSS:
\$12.9 MILLION

Poor data quality costs organizations
15-25% of annual revenue.



THE HIDDEN WORK OF DATA SCIENCE



20% Analysis & Modeling

Goal: Production-Ready
"Gold Standard" Data.

You must master the 80% to
earn the right to do the
20%.

80% Cleaning & Wrangling

PHASE 1: STRATEGIC IMPUTATION

Handle the Gaps. Don't just delete.



PHASE 2: THE INTEGRITY AUDIT

One Truth, One Record.

~~Order ID: #44902 | Product:
Nexus-X | Qty: 3 | Value:
\$1499.97 | Status: Processing |
Timestamp: 2024-01-15T14:32:21Z~~

~~Order ID: #44902 | Product: Nexus-X
Nexus-X | Qty: 3 | Value:
\$1499.97 | Status: Processing |
Timestamp: 2024-01-15T14:32:21Z~~

Order ID: #44902 | Product:
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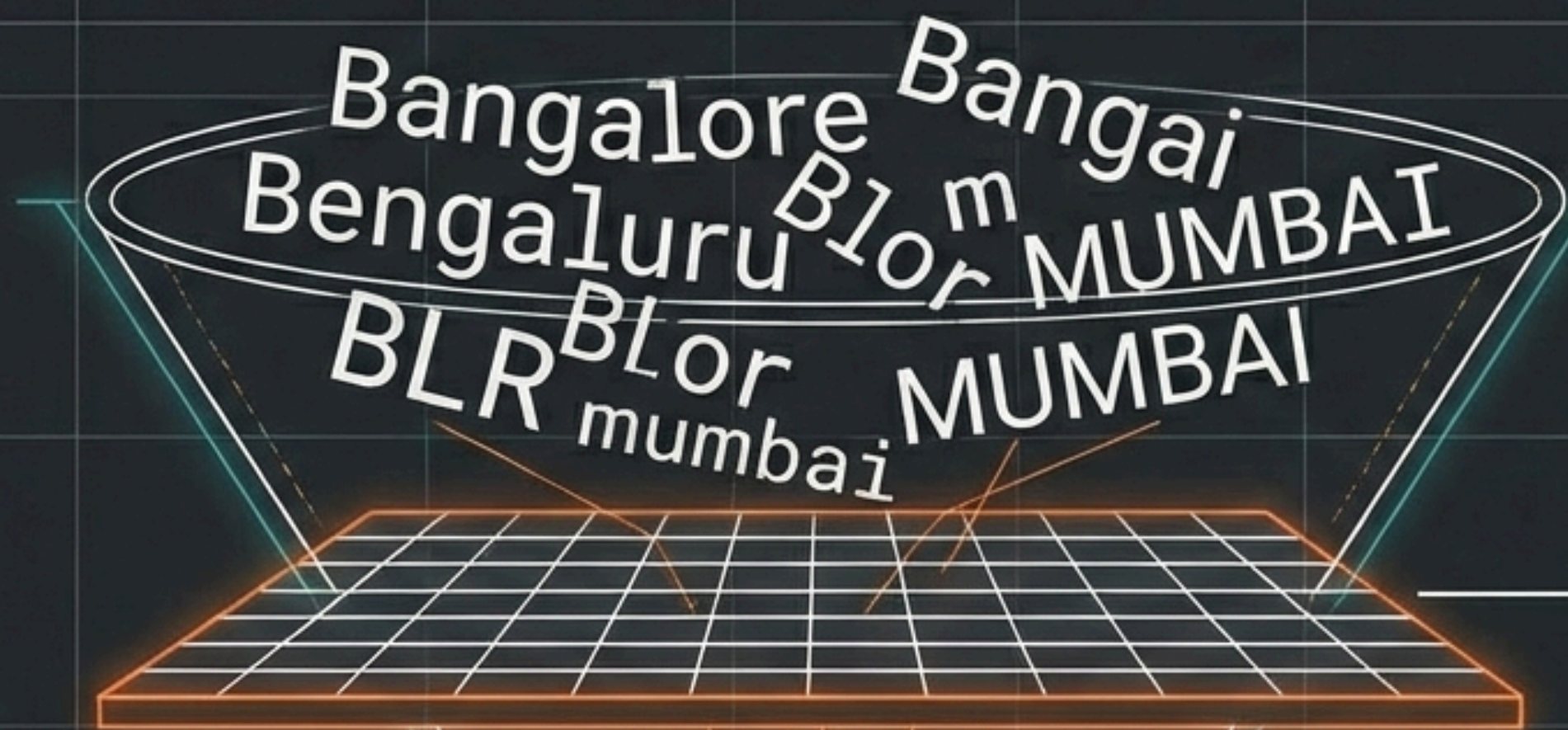
Eliminate inflated transaction counts. Unique IDs must be truly unique.

○ ○ ○

```
GROUP BY Order_ID HAVING COUNT(*) > 1
```


PHASE 3: SPEAK ONE LANGUAGE

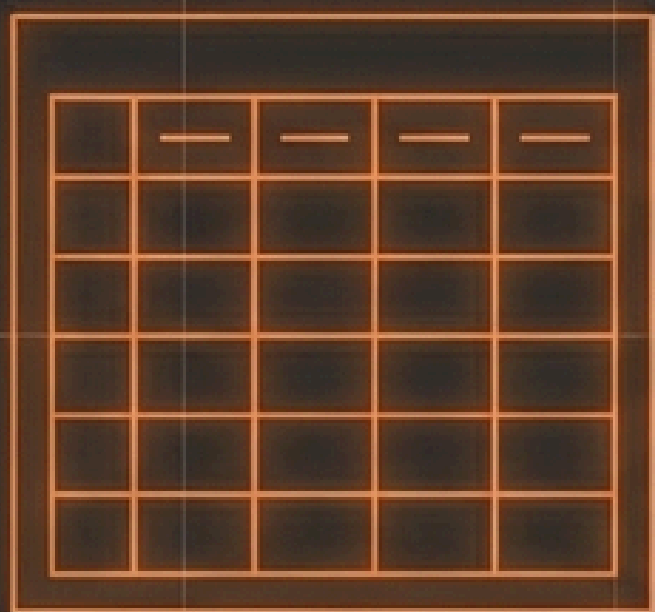
One Truth, One Record.



1. Bengaluru
2. Mumbai
3. Puducherry

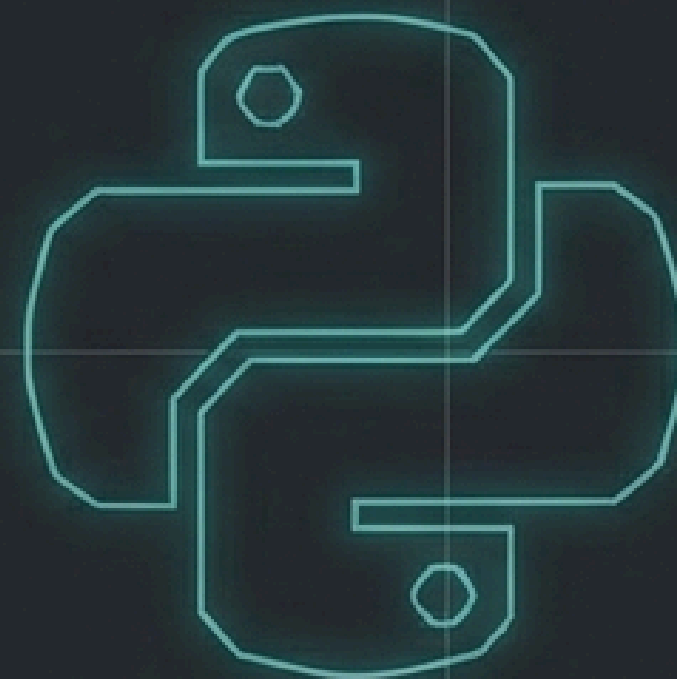
- ISO 8601 Dates (YYYY-MM-DD)
- Proper Case & Trim Whitespace
- Numeric Precision (2 decimals)
- Numeric Precision (2 decimals)

CHOOSE YOUR WEAPON: SCALE MATTERS



Excel (Power Query)

Visual. Fast.
< 1 Million Rows.



Python (Pandas)

Scalable. Automated.
> 1 Million Rows.

The Pro Move: Reproducible workflows via scripts.

If It Isn't Documented, It Didn't Happen

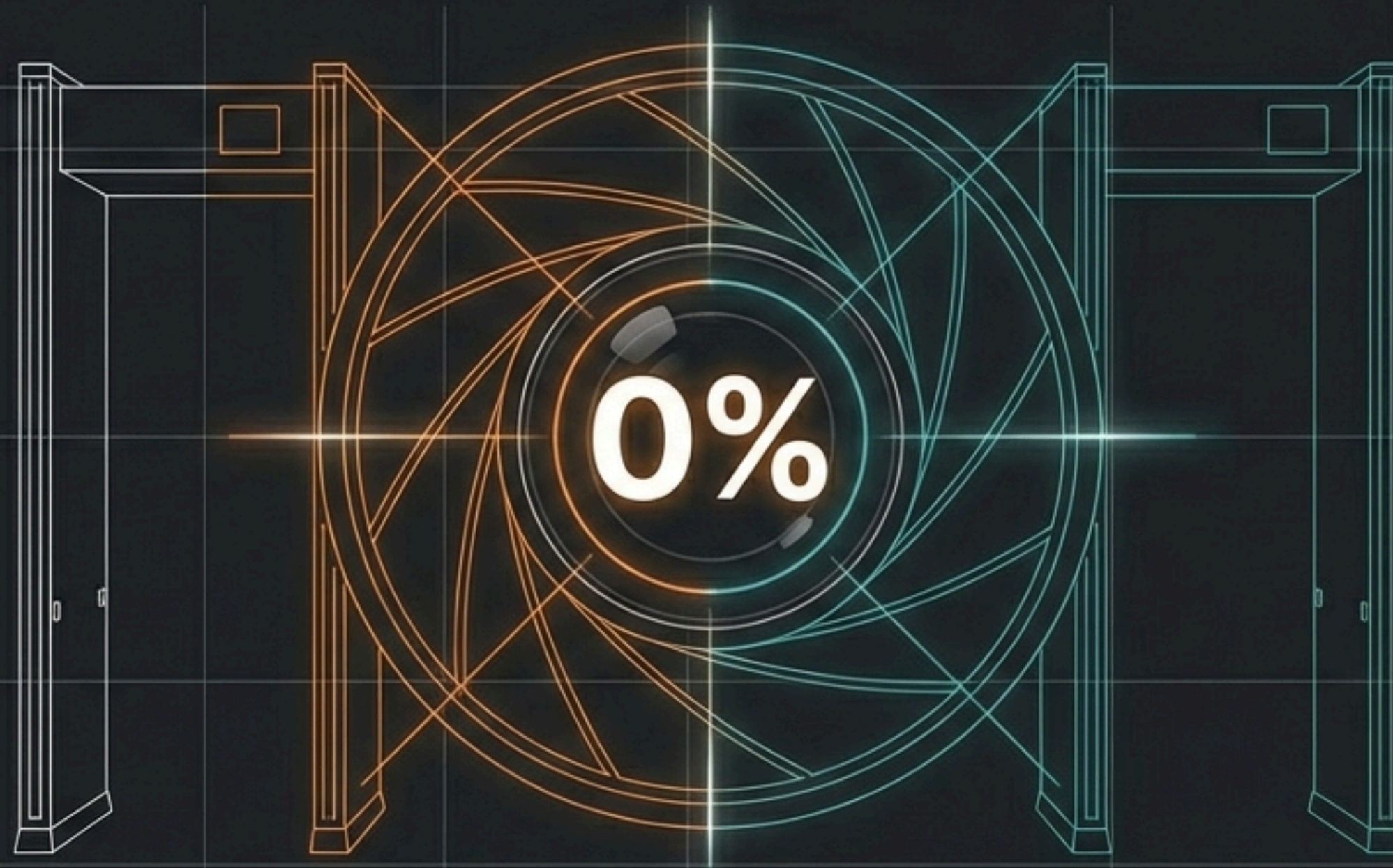
The Artifact: PDF Change Log.

Stakeholders need to know WHAT you changed and WHY.

Change ID	Description	Impact	Status
CR001	Imputed 'Age' using Median	Preserved 150 records	Resolved

THE THRESHOLD FOR PROJECT 2

Verification Gate



- 0% Error Rate on Unique Identifiers
- 0% Error Rate on Date Formats

- "Before you finish, you must prove there are zero duplicate IDs and zero incorrectly formatted dates."

From Data Consumer to Data Architect

Your analysis is only as good as your data.
Make it count.



Begin the Audit.

CONCLUSION

The absolute best way to master Data Analytics is through hands-on practice, not just theory. Don't just aim to complete these projects; take them one by one, experiment with unique solutions, and treat every "dirty" dataset as a learning opportunity. As you build these milestones at DecodeLabs, you are creating a real-world portfolio that showcases your analytical precision to future employers. Your journey to becoming a professional analyst begins right here, right now, with the very first data transformation you perform today.



THANK YOU

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 GREATER LUCKNOW, INDIA

